

ICON-LAM model

The ICON-LAM model is the limited area version on ICON ([ICO](#)sahedral [Nonhydrostatic](#)) model. Israel Meteorological Service (IMS) develops and uses the model in the framework of COSMO ([Consortium for Small-scale Modeling](#)), a consortium of the following countries: Germany, Switzerland, Italy, Greece, Poland, Romania, Russia and Israel.

IMS runs the model at [ECMWF](#) computers in the framework of [SEE-MHEWS](#) project, with the following technical features:

- Two daily runs (00Z and 12Z)
- Forecast range 90 hours.
- Model resolution is about 2.5 km (R2B10).
- Vertical resolution 65 levels, up to ~23 km.
- Deterministic [IFS](#) (ECMWF) is used as initial and boundary conditions (every 3h)
- The model domain (Fig 1.) stretches between 4-45.5E, 25.5-53N, centered at 24.75E, 39.25N and has a size of $41.5^{\circ} \times 27.5^{\circ}$ with 1,783,780 triangle cells.
- No data assimilation is currently performed

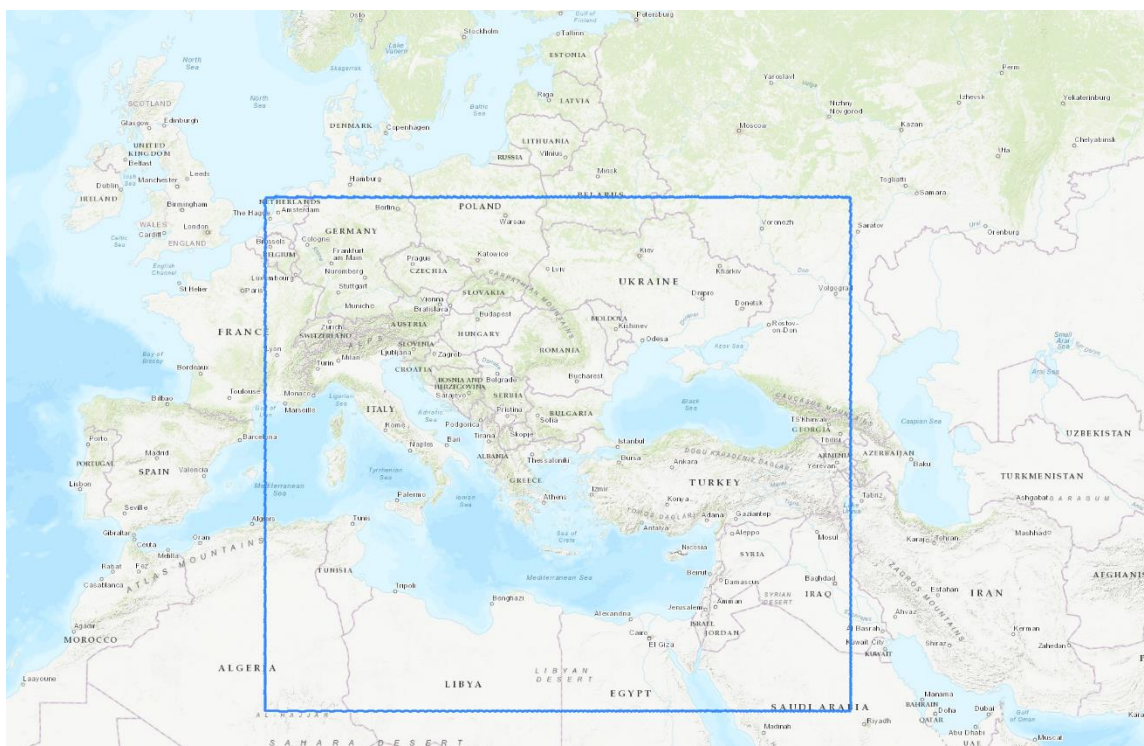


Fig 1: ICON-LAM domain

Model output

The model output over Israel is divided into NetCDF files, each file contains only one forecast field. The file name structure is IE_YYMMDDHH_"field".nc.bz2, and the options for "field" are described in table 1.

Table 1: The ICON output fields and their description

N	Field	Description	Units
*3-D fields on pressure levels			
1	geopot	geopotential at full level cell centre	m2 s-2
2	omega	vertical velocity	Pa s-1
3	rh	Relative humidity	%
4	temp	Air temperature	K
5	u	eastward wind - U-component of wind	m s-1
6	v	northward wind - V-component of wind	m s-1
Radiation fields			
7	asodifd_s	Surface down solar diff. rad. mean since model start	W m-2
8	asodifu_s	Surface up solar diff. rad. mean since model start	W m-2
9	asodird_s	Surface down solar direct rad. Mean since model start	
10	sodifd_s	shortwave diffuse downward flux at surface	W m-2
11	sob_t	Shortwave net flux at top-of-atmosphere (TOA)	W m-2
12	asob_t	TOA net solar radiation mean since model start	W m-2
13	athb_t	TOA net thermal radiation mean since model start	W m-2
Thermodynamics cloud and rain fields			
14	cape	convective available potential energy	J kg-1
15	cape_ml	cape of mean surface layer parcel	J kg-1
16	cin_ml	convective inhibition of mean surface layer parcel	J kg-1
17	hzerocl	height of 0 deg C level	m
18	clch	high level cloud, area fraction	%
19	clcl	low level cloud cover, area fraction	%
20	clcm	medium level cloud cover, area fraction	%
21	clct	total cloud cover, area fraction	%
22	hbas_con	height of convective cloud base	m
23	htop_con	height of convective cloud top	m
24	graupel_gsp	Large scale (gridscale) graupel amount	kg m-2
25	h_snow	Surface snow thickness	m
26	rain_con	convective rain	kg m-2

27	rain_gsp	Large scale (gridscale) rainfall amount	kg m-2
28	snow_con	convective snow	kg m-2
29	snow_gsp	Large scale (gridscale) snowfall amount	kg m-2
30	tot_prec	Total precipitation amount	kg m-2
31	tqv	total column integrated water vapor (precipitable water)	kg m-2
32	tqc	total column integrated cloud water	kg m-2
33	tqr	total column integrated rain	kg m-2
34	tgg	total column integrated graupel	kg m-2
35	tqs	total column integrated snow	kg m-2
36	tqi	total column integrated cloud ice content	kg m-2
Surface fields			
37	pres_msl	mean sea level pressure	Pa
38	pres_sfc	Surface air pressure	Pa
39	rh_2m	relative humidity in 2m	%
40	qv_s	specific humidity at the surface	kg kg-1
41	t_2m	temperature in 2m	K
42	td_2m	dew-point in 2m	K
43	t_g	grid mean (weighted) surface temperature	K
44	tmax_2m	Max 2m temperature	K
45	tmin_2m	Min 2m temperature	K
46	u_10m	zonal (eastward) wind component in 10m	m s-1
47	v_10m	meridional (northward) wind component in 10m	m s-1
48	gust10	Wind gust at 10 m	m s-1
49	***surf_feilds	data for T_2M, TOT_PREC, CLCT, RELHUM_2M, TD_2M, and wind	
Constant fields			
50	**topo_icon_web	ICON topography	M
Information fils			
51	***use_cams	0 – aerosol climatology 1 – CAMS forecast	-

* FI/OMEGA/RELHUM/T/U/V are 3-D fields on the following pressure levels [hPa]:
1000 975 950 925 900 875 850 825 800 750 700 650 600 500 400 350 300 250 200 150
All other fields are 2-D surface parameters

**The ICON topography files in NetCDF formats were added to MANUALS/IMS_ICON directory

***not yet implemented

***In weather and climate models, aerosols are represented by their monthly average climatology. To improve forecast on dusty and clean days, when 70% of the grid points exceed (dusty days) the AOD (Aerosol Optical Depth, which measures the extinction of a ray of light as it passes through the atmosphere) climatology by more than 0.1 or less than 0.05 (very clean days) the model uses the CAMS forecast instead of climatology.

To indicate which run was performed a file IE_YYYYMMDDHH_use_cams.txt with the value of 1 indicates the use of CAMS forecasts and 0 indicates the use of climatology.